

Review article: Numerical Methods for Engineering Quantum Computers

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ABSTRACT: In the article by Herrmann et al, “Quantum utility – definition and assessment of a practical quantum advantage,” the “Application Readiness Level” (ARL) is defined: (ARL1) “concept with unknown potential,” (ARL2) “beneficial in small idealized systems,” (ARL3) “utility indicated by theory or

resource estimations,” (ARL4) “simulated utility demonstration,” and (ARL5) “utility demonstration on quantum hardware.” At the present, practitioners have reached ARL3 with the Variational Quantum Eigenvalue (VQE) algorithm. This seems to indicate that there is still much work to be done in order to reach ARL4 or ARL5. In this review article, the latest activities from the computational “digital twin” perspective are reported on in which scientists are working hard to bring practitioners out of the Noisy Intermediate Scale Quantum (NISQ) era into an era of “Quantum supremacy.” In other words this review article discusses digital twin activities that help to advance from ARL3 to ARL4, or even ARL5. The three main categories in this review article are (1) Quantum algorithms for benchmarking quantum computers (to include emulator algorithms like Qiskit), (2) Optimization, Control, and inverse problem algorithms for Schrodinger’s equation, and (3) Error correction algorithms and algorithms for optimizing the “surface topology” of logical qubits.

The motivation for this review article is the fact that Moore’s law is no longer applicable[30]. As quoted from Shalf[38], “Moore’s Law [1] is a techno-economic model that has enabled the IT industry to double the performance and functionality of digital electronics roughly every 2 years within a fixed cost, power and area. This expectation has led to a relatively stable ecosystem (e.g. electronic design automation tools, compilers, simulators and emulators) built around general-purpose processor technologies, such as the $\times 86$, ARM and Power instruction set architectures. However, within a decade, the technological underpinnings for the process that Gordon Moore described will come to an end, as lithography gets down to atomic scale. At that point, it will be feasible to create lithographically produced devices with dimensions nearing atomic scale, where a dozen or fewer silicon atoms are present across critical device features, and will therefore represent a practical limit for implementing logic gates for digital computing [2]. Indeed, the ITRS (International Technology Roadmap for Semiconductors), which has tracked the historical improvements over the past 30 years, has projected no improvements beyond 2021, as shown in figure 1, and subsequently disbanded, having no further purpose. The classical technological driver that has underpinned Moore’s Law for the past 50 years is failing [3] and is anticipated to flatten by 2025, as shown in figure 2. Evolving technology in the absence of Moore’s Law will require an investment now in computer architecture and the basic sciences (including materials science), to study candidate replacement materials and alternative device physics to foster continued technology scaling.”

References one through three in the above quote correspond to the following references respectively: [30][28][29].

As further motivation for this review article, in Figure 3 of the article by Shalf[38], a roadmap is provided for possible paths forward when the density of circuits on classical computers exceeds a critical value. There are three categories: (a) roadmap for the next ten years, (b) 20 years, and (c) “Decades beyond exascale,” “New Models of Computation.” For category (c), Shalf[38] refers to the following prospective technology: (i) approximate computing, (ii) adiabatic reversible, (iii) Analog, (iv) Neuromorphic, and (v) quantum.

Of the “new models of computation” listed in Shalf’s article[38], this review article will address the “Numerical Methods for Engineering Quantum Computers” aspects associated with the emerging “quantum computing” paradigm. As outlined by Gamble[11], the development of reliable quantum computers will have a transformative effect on computer technology. That being said, right now we are in the “Noisy Intermediate-Scale Quantum” [6] (NISQ) era. The purpose of this review article is to bring to the forefront the current research being done, from the computational perspective, in order to move beyond the NISQ era.

1 Quantum algorithms and Quantum emulators for benchmarking quantum computers

Quantum Emulators:

1. (IBM) “Quantum Computing with Qiskit” (2024)[17]
2. “Intel Quantum Simulator: a cloud-ready high-performance simulator of quantum circuits” (2020)[13]
3. (Microsoft) “LIQUid: A Software design architecture and domain specific language for quantum computing” (2014)[42]
4. “QuESTlink—Mathematica embiggened by a hardware-optimised quantum emulator” (2020)[20]
5. (Matlab) “QCLAB: A Matlab Toolbox for Quantum Computing” (2025)[22]
6. “PennyLane-Lightning MPI: A massively scalable quantum circuit simulator based on distributed computing in CPU clusters” (2025)[21]
7. (Google) “qsim”(2020)[40]
8. “Numerical recipes in quantum information theory and quantum computing: an adventure in FORTRAN 90”(2021)[35]
9. Nguyen et al(2022)[32] “Tensor network quantum virtual machine for simulating quantum circuits at exascale”
10. Zhang et al[43], “Quantum circuit simulation with fast tensor decision diagram”

Quantum Algorithms for benchmarking Quantum Computers:

1. Callison and Chancellor (2022)[6] “Hybrid quantum-classical algorithms in the noisy intermediate-scale quantum era and beyond”
2. S. Jin, N. Liu, and Y. Yu (2024)[19], “Quantum Circuits for the heat equation with physical boundary conditions via Schrodingerisation”
3. Cerezo et al (2021)[7] “Variational quantum algorithms”

4. Lotshaw et al (2021)[27] “Simulations of frustrated Ising Hamiltonians using quantum approximate optimization”
5. Nielsen and Chuang (2010)[33] “Quantum computation and quantum information” (quantum Fourier Transform)
6. Ding and Lin (2023)[9] “Simultaneous estimation of multiple eigenvalues with short-depth quantum circuit on early fault-tolerant quantum computers”
7. Biamonte et al (2017)[3] “Quantum machine learning”
8. Benedetti et al (2019)[2] “Parameterized quantum circuits as machine learning models”
9. Khait et al(2023)[23] ”Variational quantum eigensolvers in the era of distributed quantum computers”
10. Sawaya et al (2023)[37], “HamLib: A Library of Hamiltonians for Benchmarking Quantum Algorithms and Hardware”
11. Georgadou et al(2024)[12] Hele-Shaw simulation on quantum computer.
12. Kim et al (2023)[24], “Evidence for the utility of quantum computing before fault tolerance”
13. Zoufal et al (2019)[44], “Quantum generative adversarial networks for learning and loading random distributions”
14. Nghiem and Tzu-Chieh (2023)[31], “Quantum algorithm for estimating largest eigenvalues”
15. Cadi et al (2024)[5], “Folded spectrum vqe: A quantum computing method for the calculation of molecular excited states”
16. Tilly et al (2022)[41], “The variational quantum eigensolver: a review of methods and best practices”
17. Qing and Xie (2023)[34], “Use VQE to calculate the ground energy of hydrogen molecules on IBM Quantum”
18. Ryu et al (2021)[36], “A Variational Quantum Algorithm for Ordered SVD”
19. Shirakawa et al (2024)[39], “Automatic quantum circuit encoding of a given arbitrary quantum state”
20. Harrow et al (2009)[15], “Quantum algorithm for linear systems of equations”
21. Daribayev et al (2023)[8], “Implementation of the hhl algorithm for solving the poisson equation on quantum simulators.”
22. Bravo-Prieto et al (2023)[4], ”Variational quantum linear solver”

2 Quantum Algorithms for benchmarking Quantum Computers

<https://www.ibm.com/quantum/qiskit>

Discuss the latest technologies for Quantum Computer Emulators.

Examples: (1) VQE using Slater determinant form, only valence electrons, (2) SVD (SVT, self-consistent field theory), (3) Shor's prime number factorization algorithm, (4) heat equation, and (5) Quantum Fourier Transform.

- Jin et al[18]
- Beck et al[1]

3 Quantum Algorithms with error correction logic

- Shor's algorithm for bit flip and phase flip errors. e.g. given bit flip probability p , what is the probability of a bit flip error occurring with and without error correction?
- Kubischta, Eric and Teixeira, Ian[25]
- Preskill's notes chapter 7.
- Eastin and Knill: A transversal quantum gate does not spread errors between physical qubits. "if there are n_e errors in the physical qubits representing the logical system, then after the application of a transversal gate, there will still be n_e errors." i.e., if the number of errors is below the "correctable" threshold t prior to the application of a transversal gate, it will still be correctable after the application of a transversal gate.
- Kubischta and Teixeira (a quantum code for making the 2I gates transversal (i.e. product gates))
- Ide et al[16] (surface codes)

4 Control and optimization of Quantum Computers using numerical model surrogates in place of the actual quantum computer

- Günther et al[14]
- Fei et al[10]

4.1 Control of open quantum systems

The Lindblad master equation for Open quantum systems[26] is

$$\dot{\rho}(t) = -i[H(t), \rho(t)] + \ell(\rho(t)) \quad 0 \leq t \leq T \quad (1)$$

$$\ell(\rho) = \sum_{q=1}^Q \sum_{l=1}^2 \ell_{lq} \rho \ell_{lq}^\dagger - \frac{1}{2} (\ell_{lq}^\dagger \ell_{lq} \rho + \rho \ell_{lq}^\dagger \ell_{lq}) \quad (2)$$

$$N = \prod_{q=1}^Q n_q \quad (3)$$

$$\rho(t) \in \mathbb{C}^{N \times N} \quad (4)$$

From [26], given the pure state ensemble,

$$\{q_i, |\psi\rangle_i\}_{i=1}^N, \quad (5)$$

the density operator is

$$\rho = \sum_i q_i |\psi\rangle_i \langle\psi|_i. \quad (6)$$

$$H = H_{sys} + H_c(t) \quad (7)$$

The control Hamiltonian has the form

$$H_c(t) = \sum_{q=1}^Q d_q(\alpha_q, t) a_q + d_q(\alpha_q, t)^*(t) a_q^\dagger. \quad (8)$$

The objective function has the form

$$\mathcal{G}(\alpha) = \frac{1}{M} \sum_{i=1}^M \left(\beta_i J(\rho_i^{target}, \rho_i(T)) + \gamma_1 \int_0^T \beta(t) J(\rho_i^{target}, \rho_i(t)) dt \right) + \gamma_2 \|\alpha\|_2^2 / 2 \quad (9)$$

In Quandary the following choices for J are[14]

$$J_{Frob}(\rho^{target}, \rho) = \|\rho^{target} - \rho\|_F^2 / 2 \quad (10)$$

$$J_{Trace}(\rho^{target}, \rho) = 1 - Tr((\rho^{target})^\dagger \rho) \quad (11)$$

$$J_{N_m}(\rho) = Tr(N_m \rho) \quad (12)$$

4.2 Switching time optimization Control of binary quantum systems

$$\dot{X}(t) = -iH(t)X(t) \quad 0 \leq t \leq T \quad (13)$$

$$H(t) = H^{(0)} + \sum_{j=1}^J u_j(t)H^{(j)} \quad 0 \leq t \leq T \quad (14)$$

$$\sum_{j=1}^J u_j(t) = 1 \quad 0 \leq t \leq T \quad (15)$$

$$u_j(t) \in \{0, 1\} \quad j = 1, \dots, J \quad (16)$$

Cost function example ($Q = 2$):

$$\mathcal{G}(X[u_1(t), \dots, u_J(t)](T)) = 1 - \frac{1}{4} |\text{Trace} \{ X_{CNOT}^\dagger X(T) \}| \quad (17)$$

- compare VQE on classical computer vs IBM hardware to demonstrate the need for improved hardware.
- compare QUANDARY to other quantum optimal control algorithms. inverse problems.
- compare Error correction algorithms.
- quantum and hybrid classical/quantum algorithms for benchmarking quantum computers.

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